

9.1 Secure Networks with Firewalls

Secure Networks with Firewalls

Firewalls

All firewalls share some common properties:

- Firewalls are resistant to network attacks.
- Firewalls are the only transit point between internal corporate networks and external networks because all traffic flows through the firewall.
- Firewalls enforce the access control policy.

Different types of firewalls have different benefits and limitations.

Allow traffic from any external address to the web server.

Allow traffic to FTP server.

Allow traffic to SMTP server.

Allow traffic to internal IMAP server.

Deny all inbound traffic with network addresses matching internal-registered IP addresses.

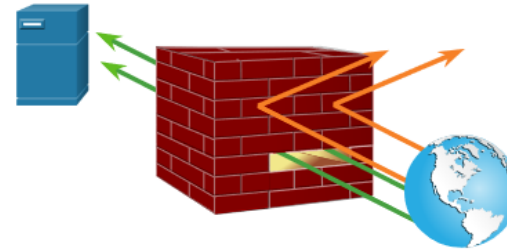
Deny all inbound traffic to server from external addresses.

Deny all inbound ICMP echo request traffic.

Deny all inbound MS Active Directory queries.

Deny all inbound traffic to MS SQL server queries.

Deny all MS Domain Local Broadcasts.



Types of Firewalls

It is important to understand the different types of firewalls and their specific capabilities so that the right firewall is used for each situation.

- **Packet Filtering (Stateless) firewalls** - These are usually part of a router firewall, which permits or denies traffic based on Layer 3 and Layer 4 information. They are stateless firewalls that use a simple policy table look-up that filters traffic based on specific criteria.
- **Stateful firewalls** – These are the most versatile and the most common firewall technologies in use. Stateful firewalls provide stateful packet filtering by using connection information maintained in a state table. Stateful filtering is a firewall architecture at the network layer.
- **Application Gateway (Proxy) firewalls** – These filter information at Layers 3, 4, 5, and 7. Most of the firewall control and filtering is done in software. When a client needs to access a remote server, it connects to a proxy server.
- **Next Generation firewalls** – These go beyond stateful firewalls by providing integrated intrusion prevention, application awareness and control, upgrade paths to include future information feeds, and techniques to address evolving security threats.

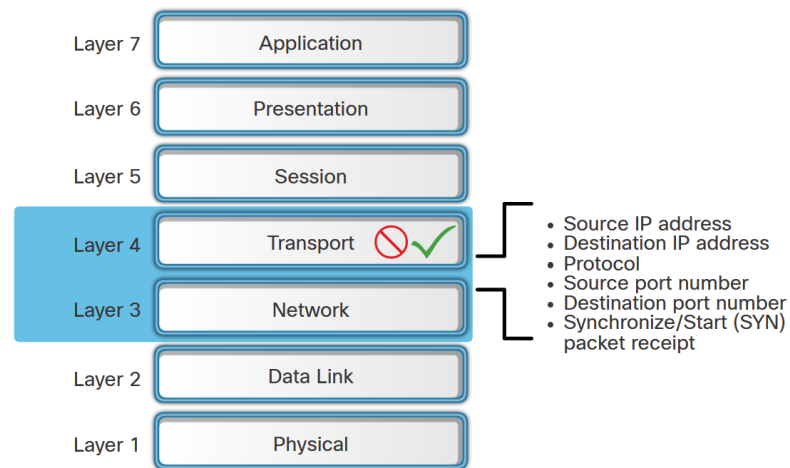
Packet Filtering Firewall Benefits and Limitations

Benefits of Packet Filtering Firewalls:

- They implement simple permit or deny rule sets.
- They have a low impact on network performance.
- They are easy to implement and are supported by most routers.
- They provide an initial degree of security at the network layer.
- They perform many of the tasks of a high-end firewall at a much lower cost.

Limitations of Packet Filtering Firewalls:

- They are susceptible to IP spoofing.
- They do not reliably filter fragmented packets.
- They use complex ACLs, which can be difficult to implement and maintain.
- They cannot dynamically filter certain services.
- They are stateless.



Stateful Firewall Benefits and Limitations

The table lists benefits and limitations of stateful firewalls.

Benefits	Limitations
Primary means of defense	No Application Layer inspection
Strong packet filtering	Limited tracking of stateless protocols
Improved performance over packet filters	Difficult to defend against dynamic port negotiation
Defends against spoofing and DoS attacks	No authentication support
Richer data log	

9.2 Firewalls in Network Design

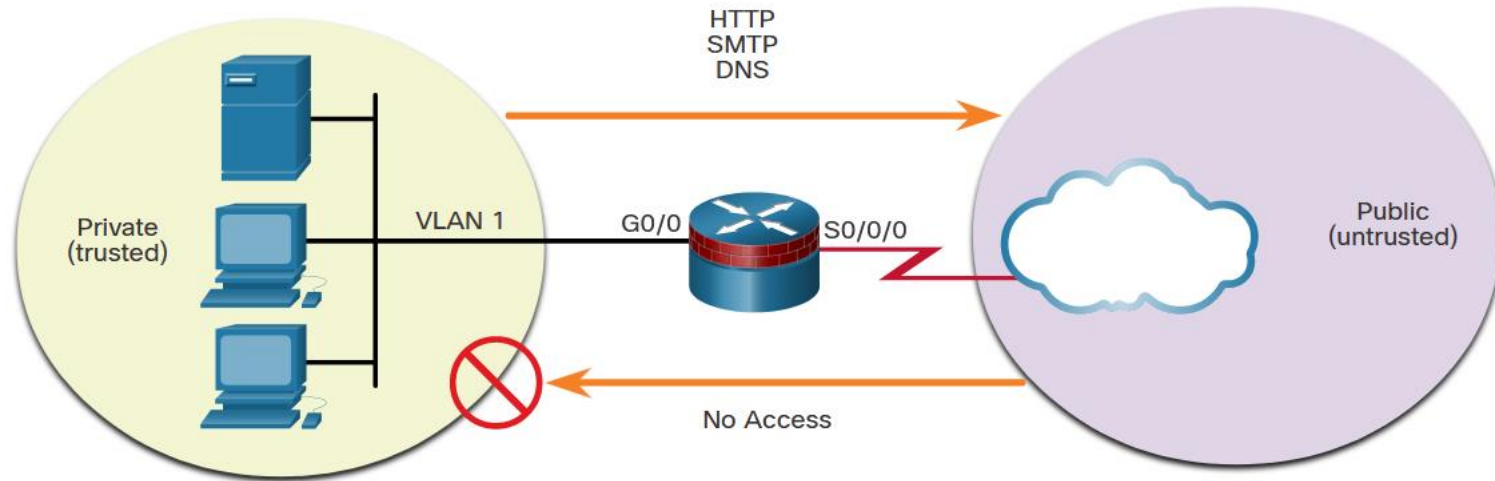
Common Security Architectures

Firewall design is primarily about device interfaces permitting or denying traffic based on the source, the destination, and the type of traffic. Here are three common firewall designs:

- **Private and Public**
- **Demilitarized Zone (DMZ)**
- **Zone-Based Policy**

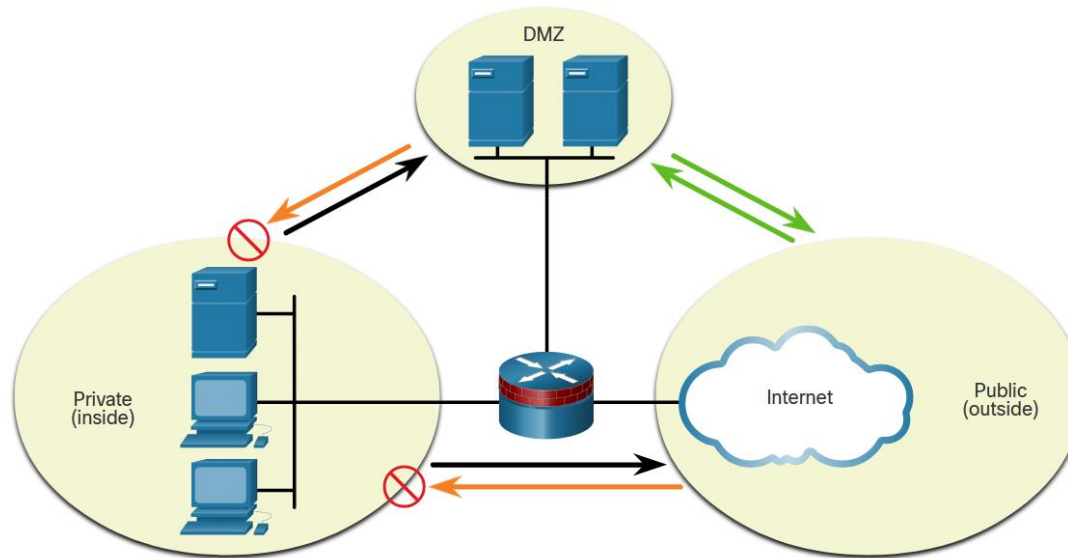
Common Security Architectures (Cont.)

Private and Public - The public network (or outside network) is untrusted, and the private network (or inside network) is trusted.



Common Security Architectures (Cont.)

Demilitarized Zone (DMZ) - This is a firewall design where there is typically one inside interface connected to the private network, one outside interface connected to the public network, and one DMZ interface.



Legend

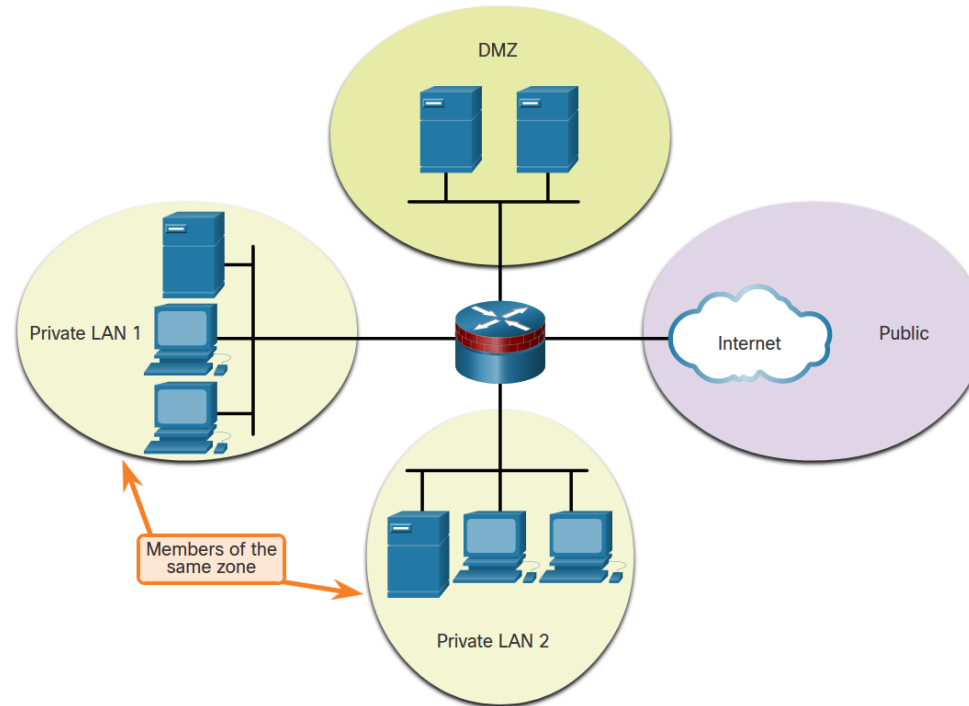
— Selectively permitted

— Blocked

— Inspected and permitted with little or no restriction

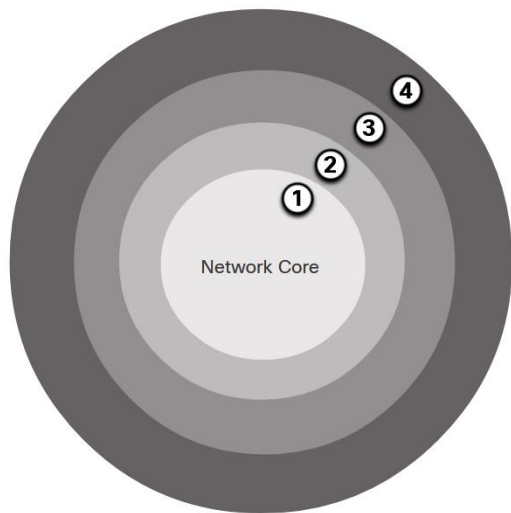
Common Security Architectures

Zone-Based Policy - Zone-based policy firewalls (ZPFs) use the concept of zones to provide additional flexibility. A zone is a group of one or more interfaces that have similar functions or features. Zones help you specify where a Cisco IOS firewall rule or policy should be applied.



Layered Defense

A layered defense uses different types of firewalls that are combined in layers to add depth to the security of an organization. Policies can be enforced between the layers and inside the layers. The following example shows four layers of security.



- 1. Network Core security** - Protects against malicious software and traffic anomalies, enforces network policies, and ensures survivability
- 2. Perimeter security** - Secures boundaries between zones
- 3. Communications security** - Provides information assurance
- 4. Endpoint security** - Provides identity and device security policy compliance

Layered Defense (Cont.)

This partial list of best practices can serve as a starting point for a firewall security policy.

- Position firewalls at security boundaries. Firewalls are a critical part of network security, but it is unwise to rely exclusively on a firewall for security.
- Deny all traffic by default.
- Permit only services that are needed.
- Ensure that physical access to the firewall is controlled.
- Regularly monitor firewall logs.
- Practice change management for firewall configuration changes.
- Remember that firewalls primarily protect from technical attacks originating from the outside.

